Universal spectral freedom for a fixed positive kernel

Autonomous proof attempt for arXiv:2104.11807

11 August 2026

Abstract

Question 2.12 of Jorgensen–Song–Tian asks for the spectrum of $T_\mu T_\mu^*$ as the regular measure μ varies for a fixed positive-definite kernel. We exhibit one fixed elementary kernel for which the answer is maximally free: every nonempty closed subset of $[0,\infty)$ occurs, including unbounded sets and sets with isolated zero. We give the closed-operator domains explicitly. Consequently no kernel-independent restriction stronger than positivity and closedness is possible.

1 The theorem

For a p.d. kernel K on X, the source defines $\mu \in \mathrm{Reg}(K)$ when $K(\cdot, x) \in L_2(\mu)$ for every x, and defines T_μ initially as the identity inclusion of the span of the kernel sections into $L_2(\mu)$. The operator is closable; below T_μ denotes its closure. The source asks for $\sigma(T_\mu T_\mu^*)$ as μ varies [1, Question 2.12].

Let $X = \{0\} \cup \mathbb{N}$ and fix

$$K(0,x) = K(x,0) = 0, \qquad K(n,m) = \delta_{nm} \quad (n,m \in \mathbb{N}). \tag{1}$$

This is positive definite. Its kernel sections are $0, e_1, e_2, \ldots$, so they are pairwise distinct and the source distance $d_K(x,y) = \|K_x - K_y\|$ is a genuine metric.

Theorem 1 (Universal fixed kernel)**.** *For the single kernel* (1),

$$\{\sigma(T_\mu T_\mu^*) : 0 \neq \mu \in \mathrm{Reg}(K)\} = \{S \subset [0,\infty) : S \text{ is nonempty and closed}\}.$$

The inclusion from left to right is automatic because $T_\mu T_\mu^*$ is positive self-adjoint. The content is the converse, proved below.

2 Exact diagonalization, including domains

The RKHS of (1) is

$$\mathcal{H}_K = \{f : X \to \mathbb{C} : f(0) = 0, (f(n))_{n\geq 1} \in \ell_2\}, \qquad \|f\|^2_{\mathcal{H}_K} = \sum_{n\geq 1} |f(n)|^2.$$

Every Borel measure on this discrete X has weights $a = \mu(\{0\})$ and $w_n = \mu(\{n\})$. It lies in $\mathrm{Reg}(K)$ exactly when each $w_n < \infty$, because $\|K_n\|^2_{L_2(\mu)} = w_n$; the mass a is irrelevant to regularity. We use nonzero measures with finite singleton weights, hence they are σ-finite.

Discard the indices with $w_n = 0$ from $L_2(\mu)$. The unitary

$$V : L_2(\mu) \longrightarrow \big(\mathbb{C} \text{ if } a > 0\big) \oplus \ell_2(\{n : w_n > 0\}), \qquad Vf = \big(\sqrt{a} f(0), (\sqrt{w_n} f(n))_n\big)$$

turns T_μ on finite sequences into

$$c = (c_n) \longmapsto \big(0, (\sqrt{w_n} c_n)_n\big).$$

Its closure and adjoint therefore have the standard diagonal domains

$$\operatorname{dom}(T_\mu) = \{c \in \ell_2 : \sum_n w_n |c_n|^2 < \infty\},$$
$$\operatorname{dom}(T_\mu^*) = \{(z, y) : \sum_n w_n |y_n|^2 < \infty\}, \qquad T_\mu^*(z, y) = (\sqrt{w_n} y_n)_n.$$

Consequently

$$\begin{aligned} V(T_\mu T_\mu^*)V^{-1} &= 0 \oplus M_w, \qquad\qquad M_w(y_n) = (w_n y_n), \\ \operatorname{dom}(M_w) &= \{y : \sum_n w_n^2 |y_n|^2 < \infty\}. \end{aligned} \tag{2}$$

where the zero summand is present exactly when $a > 0$.

Lemma 2. *For the atomic measure above,*

$$\sigma(T_\mu T_\mu^*) = \overline{\{w_n : w_n > 0\}} \ \cup \ \begin{cases} \{0\}, & a > 0, \\ \varnothing, & a = 0. \end{cases}$$

Proof. Equation (2) reduces the claim to a diagonal self-adjoint operator, even when the weights are unbounded. A complex number λ is in its resolvent precisely when $\inf_n |w_n - \lambda| > 0$; then the inverse is the bounded diagonal operator with entries $(w_n - \lambda)^{-1}$. This proves the closure formula. The extra coordinate contributes the stated zero eigenvalue. □

The same answer follows directly from the source integral formula:

$$(T_\mu T_\mu^* f)(n) = \sum_{m \in X} K(m, n) f(m) \mu(\{m\}) = w_n f(n), \qquad (T_\mu T_\mu^* f)(0) = 0,$$

with (2) supplying the required maximal domain.

3 Realizing an arbitrary closed set

Let $S \subset [0, \infty)$ be nonempty and closed. If $S_+ = S \cap (0, \infty)$ is nonempty, choose a sequence (s_n) in S_+ whose closure, together with 0 when $0 \in S$, is S; repeat elements when S_+ is finite. Set $w_n = s_n$. If $S_+ = \varnothing$, set every $w_n = 0$. Finally put

$$\mu(\{0\}) = \begin{cases} 1, & 0 \in S, \\ 0, & 0 \notin S. \end{cases}$$

All singleton weights are finite, so $\mu \in \operatorname{Reg}(K)$, and $\mu \neq 0$. Lemma 2 gives $\sigma(T_\mu T_\mu^*) = S$, proving Theorem 1.

Remark 3 (Scope)**.** *The theorem completely answers the variation problem for the fixed kernel* (1) *and classifies what can happen without assumptions on K. It does not give a closed-form classification for each arbitrary fixed kernel. The latter necessarily depends on that kernel's integral quadratic form; the example shows that no further universal spectral law can exist.*

References

[1] P. E. T. Jorgensen, M.-S. Song, and J. Tian, *Positive Definite Kernels, Algorithms, Frames, and Approximations*, arXiv:2104.11807v1 (2021).